

NUSRAT ASRAFI

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SUMMARY

IBM Certified Entry-level Data Scientist with five years of research experience in data analysis, modeling, reporting, dashboard, and implementation in cyber security data. Proven understanding of predictive analytics, statistical analysis, and machine learning, Data pipelines with Python, SQL, and Plotly Dash applications. Passionate about developing and improving models to solve real-world problems.

RESEARCH INTEREST

Machine Learning, ETL (Extract Transform Load), Predictive Modeling, Statistics

EDUCATION

Master of Science in Computer Science Kennesaw State University, Marietta, Georgia	<i>Jan 2017-May 2020</i>
Certificate in Computer Science Foundation Kennesaw State University, Marietta, Georgia	<i>Jan 2017-Dec 2018</i>
Bachelor of Science in Electrical & Electronics Engineering Ahsanullah University of Science and Technology, Dhaka, Bangladesh	<i>May 2010-June 2014</i>

WORK EXPERIENCE

Research Volunteer , Department of Applied Statistics Kennesaw State University, Marietta, GA - Analyzed blood pressure characteristics of child and women data with Box-Cox transformation using Python with five team members	<i>Oct 2020- May 2021</i>
Graduate Research Assistant , Department of Computer Science Kennesaw State University, Marietta, GA - Conducted research on static and dynamic analysis of malware on android apps and computer devices - Preprocessing and analyzing the data with machine learning algorithms to predict the future behavior of malware	<i>Jan 2017 – Dec 2018</i>
Teaching Assistant , Department of Computer Science Kennesaw State University, Marietta, GA - Managed grading in the Introduction to Programming (C++) Lab in the Department of Computer Science	<i>Jan 2019-May 2019</i>

AWARDS AND CERTIFICATIONS

Professional Certificate: Springboard Data Science Career Track IBM Data Science Professional Certificate	<i>Ongoing</i> <i>Nov 2022</i>
Awards: Best Abstract Paper Awards By ACM Conference 2020 Deans 4.0 club Awards, Kennesaw State University	<i>April 2020</i> <i>Fall 2017</i>

PEER-REVIEWED CONFERENCE PUBLICATIONS

[C1] Asrafi, N., Lo, D.C.T., Parizi, R.M., Shi, Y. and Chen, Y.W., 2020, April. Comparing Performance of Malware Classification on Automated Stacking. In Proceedings of the 2020 ACM Southeast Conference (pp. 307-308). (**Best Abstract Paper**) [DOI]

[C2] Asrafi, N., 2019, April. Comparing Performances of Graph Mining Algorithms to Detect Malware. In Proceedings of the 2019 ACM Southeast Conference (pp. 268-269). [DOI]

MASTER'S THESIS

Performance of Malware Classification with Machine Learning using Feature Selection

- The project aimed to classify malware families using feature selection and machine learning algorithms.
- Build a malware family classifier with 82 percent accuracy from the frequency of API calls (limited to first 100 API calls) using ML pipelines
- Tools Used: Python
- Machine Learning Algorithms: MLP, Random Forest, and SVM with linear, RBF, and polynomial kernel
- Feature Selection: FDR-ANOVA, RFE, LASSO, FDR-Chi2

PROJECTS

IBM Data Science Caption Project

- Cost Estimation of Falcon Spaceship Project- Reusing the first stage of the spaceship is the key to making a cost-effective spaceship. Find out the several factors to safely landing the spaceship for reusing the first stage. [Project GitHub]

Tools used: Python, SQL, interactive dashboard, Microsoft PowerPoint

Heart Disease Diagnosis using machine learning algorithm:

- Time Complexity Analysis of ML pipeline using heart diagnosis data set: In terms of time complexity, Logistic Regression performed best in terms of area under the receiver operating characteristic curve (AUC-ROC) and SVM took the most time in the training phase.

Tools used: Python

Machine Learning Algorithms: Decision Tree, Random Forest, Naive Bayes, Logistic Regression, and Support Vector Machine

Toxic Comment Challenges:

- Predict the probability of toxicity from multi-class comments of 56.32 MB Wikipedia comments using two regression models where Logistic Regression performs the best in terms of AUC-ROC and accuracy

Tools used: Python

Machine Learning Algorithms: Random forest, Logistic Regression

RELATED COURSEWORK

Big Data Analysis, Machine Learning, Machine Learning for Business Professionals (Coursera), Data Structure & Algorithms, Advanced Algorithms, Computer Architecture & Operating Systems, and Foundation of Programming (Java).

TECHNICAL SKILLS

Programming Languages: Python, SQL, JAVA, C++

Libraries for Data Science: Python (Pandas, NumPy, Matplotlib, Seaborn, Folium, Scikit-learn, Scipy)

Data Visualization: Tableau, Power BI, Plotly Dash

Cloud Platform: IBM Watson Studio, IBM db2, IBM SPSS

Miscellaneous Technologies: Excel, Git

LEADERSHIP AND PROFESSIONAL SERVICE

- Worked as IT Director of Graduate Student Association (GSA) at Kennesaw State University (KSU).
- Successfully organized a welcome event for new students of GSA in Spring 2018.
- Member of KSU Robotics Club.
- Class representative at Ahsanullah University of Science and Technology (AUST).
- Member of ACMSE in 2020.